

Course Framework

Program Name	Full Stack Web Development
Course Name	Version Control with Git and GitHub
Name of Educational Developer	Angela Tessier
Name of SME(s)	Naresh Nagandla
Course Hours	20 hrs
Course Delivery	Online Instructor-Guided
Course Description	The Version Control with Git and Github course will provide you with a comprehensive, hands-on foundation of the Git version control system. Effective use of version control is an important and useful skill for any developer working on multi-developer projects. Topics include installation and setup of Git, commits, diffs, logs, cloning, push, pull, branching and merging, and resolving conflicts, and Github.
Course Rationale	A course on version control and GitHub is crucial for anyone involved in software development. It not only addresses the technical aspects of code management but also instills best practices for collaboration, code quality, and project management.

Course Outcomes and Sub Outcomes	Learning Outcomes Upon successful completion of this course students will be able to: 1. Demonstrate how to Install and configure Git for use 2. Use Git to manage files using CLI commands 3. Discuss the use cases and benefits of GitHub Sub Outcomes Outcome 1: Demonstrate how to Install and configure Git for use • Demonstrate how to install and configure Git locally • Demonstrate how to initialize a local repository • Create an initial commit • Explain how to view a diff between commits • Demonstrate how to add a remote repo • Create a README file with the repository description • Demonstrate how to commit and push changes • Create rules to ignore files on commit • Discuss how to rebase of a repository • Discuss how to clone of a repository • Demonstrate how to resolve conflicts Outcome 2: Use Git to manage files using CLI commands • Use CLI commands to create, clone and manage repositories • Use CLI commands to perform branching, merging and rebasing • Describe how to prevent and resolve merge conflicts • Describe common Git workflows
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	• Discuss basic troubleshooting of Git Outcome 3: Discuss the use cases and benefits of GitHub • Contrast the differences between Git and Github about their use and purpose • Explain how to sign up for Github to organize repositories • Explain how to host a simple website on Github pages
Overall Assessment Scheme	Assignments- 75% Quiz - 25%
Assessment Breakdown	Assignment 1: Getting Started with Git and GitHub - 30% of overall grade Assignment 2: Create new Branch, merge and Collaborate changes with GitHub - 30% of overall grade Assignment 3: Cloning a GitHub Repository of your choice - 15% of overall grade Quiz on Version Control with Git and GitHub - 25% of overall grade
Required Textbook	Hartl, M. (2020). Learn enough git to be dangerous: An introduction to version control. Open Source.
Required Courseware (from Publisher)	N/A
Required Software (Not from Publisher)	N/A
Required Equipment	N/A

Assessment Scheme

Assessment	Description	Weight
Assignment 1: Getting Started with Git and GitHub	Students will create a local git repository and create new HTML files and make a first commit and then push it to GitHub	30%
Assignment 2: Create new Branch, merge and collaborate changes with GitHub	Students will create a new branch and merge changes, also perform pull and push the changes to GitHub.	30%
Assignment 3: Cloning a GitHub Repository of your choice	Students will choose an existing GitHub repository of their choice and clone into local repository and explore the logs with in local repository	15%

Quiz on Version control with Git and GitHub	Students will need to answer the multiple choice questions which are based on the textbook chapters 1-4.	25%
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Assessment Mapping

Assessment	Outcome 1: Demonstrate how to Install and configure Git for use	Outcome 2: Use Git to manage files using CLI commands	Outcome 3: Discuss the use cases and benefits of GitHub
Assignment 1: Version Control with Git and GitHub	x		x
Assignment 2: Version Control with Git and GitHub		x	x
Assignment 3: Cloning a GitHub Repository of your choice			x
Quiz on Version control with Git and GitHub	x	x	x

Module/ Topic			
Approximate time (hours)	Course Learning Outcome and Sub-outcomes	Summative Assessments (include title, description and grade points/ weightings)	Activities (Facilitated, Peer-to-Peer & Independent) *Include Hyperlinks to Textbook Chapters, Resources and Materials
Essential Questions			
Module1 Version Control with Git and GitHub 20 Hrs	Learning Outcomes Upon successful completion of this course students will be able to: 1. Demonstrate how to Install and configure Git for use 2. Use Git to manage files	Assignment 1:  Assignment 1 Getting Start... Weight: 30% Assignment 2  Assignment 2 Create new ... Weight: 30% Assignment 3:  Assignment 3 Cloning a Gi... Weight : 15%	Module 1 Content:  Version Control with Git and Github.pptx External Videos : Title: What is Version Control LearnWebCode. (2016, October 18). <i>Git Tutorial Part 1:</i> <i>What is Version Control?</i> [Video]. YouTube. https://www.youtube.com/watch?v=9GKpblIsiow

	using CLI commands 3. Discuss the use cases and benefits of GitHub Outcome 1: Demonstrate how to Install and configure Git for use Sub-Outcomes: • Demonstrate how to install and configure Git locally • Demonstrate how to initialize a local repository • Create an initial commit • Explain how to view a diff	Quiz : W Quiz on Version Control wit... Weight : 25%	Youtube Link: 📺 Git Tutorial Part 1: What is Version Control? Watch the video What is Version Control? After viewing, answer the following questions: 1. What is Git? 2. What does Git do? 3. Why should you use Git? List any questions that you may have. Post a minimum of one question and discuss with your instructor and classmates through the Help Forum. Title: Git & GitHub for Beginners Traversy Media. (2017, February 5). <i>Git & GitHub crash course for beginners</i> [Video]. YouTube. https://www.youtube.com/watch?v=SWYqp7iY_Tc Youtube Link : 📺 Git & GitHub Crash Course For Beginners Watch the video Git & Github for Beginners. Take note of the basic and most important commands like add, commit, status, push, etc. List any question that you have
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	between commits • Demonstrate how to add a remote repo • Create a README file with the repository description • Demonstrate how to commit and push changes • Create rules to ignore files on commit • Discuss how to rebase of a repository • Discuss how to		about the topics in the video. Post a minimum of one question and discuss with your instructor and classmates through the Help Forum. Title : Using Git with Visual Studio TechGenix. (2017, April 7). <i>Using Git with Visual Studio</i> [Video]. YouTube. https://www.youtube.com/watch?v=jUiulAZt6Dw Youtube Link :  Using Git with Visual Studio View the video Git with Visual Studio. Summarize the points shown in the video, like creating a new project in Git, connecting an existing project to Git, clone a project from a remote Git repository, etc. Note the other relevant topics in the video, and list any questions you have. Post a minimum of one question and participate in discussions with your instructor and classmates through the Help Forum.
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	clone of a repository • Demonstrate how to resolve conflicts Outcome 2: Use Git to manage files using CLI commands Sub-outcomes: • Use CLI commands to create, clone and manage repositories • Use CLI commands to perform branching, merging and rebasing • Describe how to prevent		
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	and resolve merge conflicts • Describe common Git workflows • Discuss basic troubleshooting of Git Outcome 3: Discuss the use cases and benefits of GitHub Sub-Outcomes: • Contrast the differences between Git and Github about their use and purpose • Explain how to		
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	sign up for Github to organize repositories • Explain how to host a simple website on Github pages		
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